Practical No: 7-B

Aim: Develop an Android application demonstrating Event Handling using Listeners (OnClick, OnLongClick, etc.).

Event Handling

Event Handling is the process of responding to user actions such as:

- Button Click
- Long Press
- Touch Events
- Item Selection

Android provides Listeners to handle these events.

Common Listeners

- `OnClickListener` → **Triggered when a view is clicked.**
- `OnLongClickListener` → **Triggered when a view is pressed and held.**

acivity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity"
   android:orientation="vertical"
   android:gravity="center">

   <Button
       android:id="@+id/btnClick"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Click Me"/>

   <Button
       android:id="@+id/btnLongClick"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Long Click Me"
       android:layout_marginTop="20dp"/>
```

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```
</LinearLayout>
```

MainActivity.java

```
package com.example.pract1;

import android.annotation.SuppressLint;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.MotionEvent;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   Button btnClick, btnLongClick;

   @SuppressLint("MissingInflatedId")
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       EdgeToEdge.enable(this);
       setContentView(R.layout.activity_main);
       btnClick = findViewById(R.id.btnClick);
       btnLongClick = findViewById(R.id.btnLongClick);

       // OnClick Listener
       btnClick.setOnClickListener(v -> {

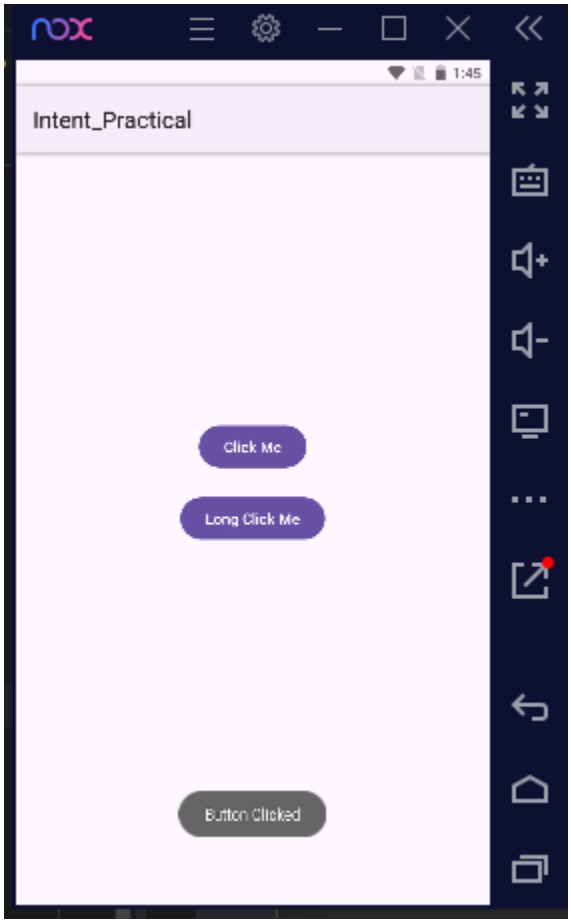
           Toast.makeText(
                   MainActivity.this,
                   "Button Clicked",
                   Toast.LENGTH_SHORT
           ).show();

       });

       // OnLongClick Listener
       btnLongClick.setOnLongClickListener(v -> {
```

```
            Toast.makeText(
                    MainActivity.this,
                    "Button Long Pressed",
                    Toast.LENGTH_SHORT
            ).show();
            return true;
        });
    }
}
```

Output:



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